

Intro to Laser Engraving

Jon A 5/2/26



Objectives

- Learn basic Inkscape design
- Importing a design for cutting
- Learn to operate under supervision by an authorized person
- Safe Operation



Requirements

- Must be supervised by authorized user
- Must use only during office hours or open-house.
- 18+ or supervised by parent/guardian
- Donation to Spark
 - \$40 Non-members
 - \$20 Members
 - Please help us, it takes time and effort to keep these machines in good condition.
 - We also have to pay for facilities, electricity, etc.
 - You will not find a better deal anywhere!



Next Steps

- After this training, feel free to come to the space during open-hours and use the machines provided that an authorized trainer is available to supervise machine operation. You will have to provide your own designs and materials, but use of the machines is free.
- If you become a member, there is additional training to:
 - Operate unsupervised or during non open-house hours.
 - Maintain and align machine
- If you enjoy this, we encourage you to become a member and get actively involved in the community.
- Tell your friends how much fun it is!



Topics Covered

- Machine Basics
- We will go through a basic design in Inkscape
- We will import file into Lightburn for cutting
- We will set up cutter for operation
- You will laser etch your own cutting board to take home

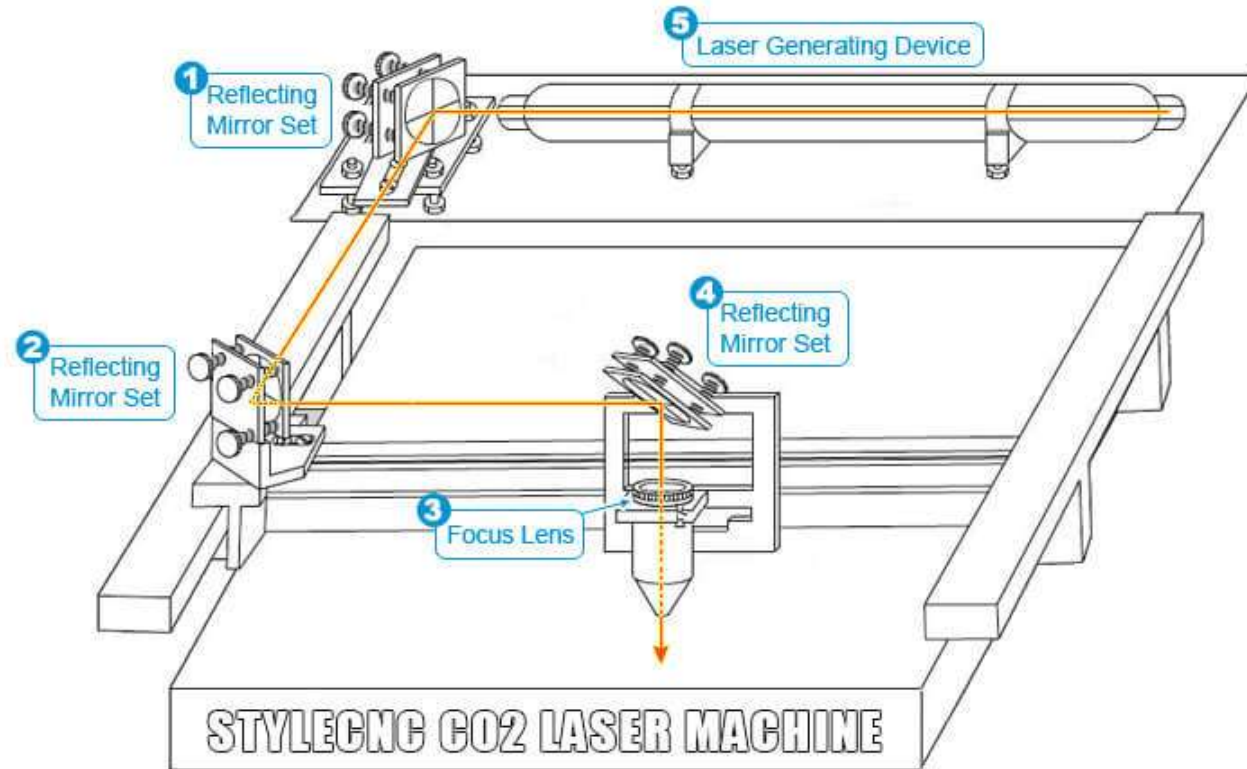


Machine Basics

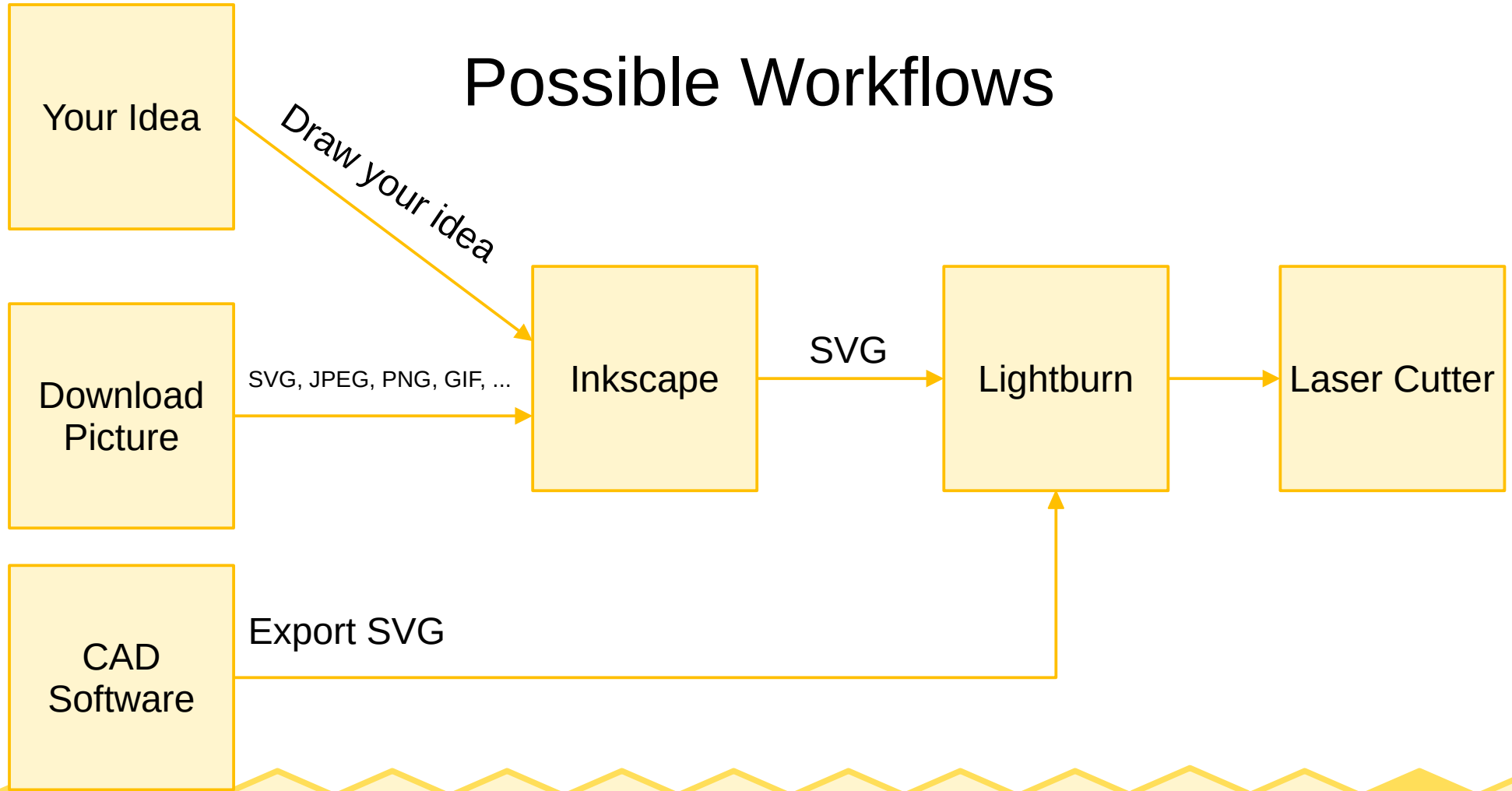
- 150 Watt CO2 Laser (10.6um wavelength)
- Water cooled
- Adjustable bed height
- Usable Work Area: 550mm x 850mm (21.5in by 33.4in)
- Air extraction
- Air assist to improve cut quality/move debris away from nozzle



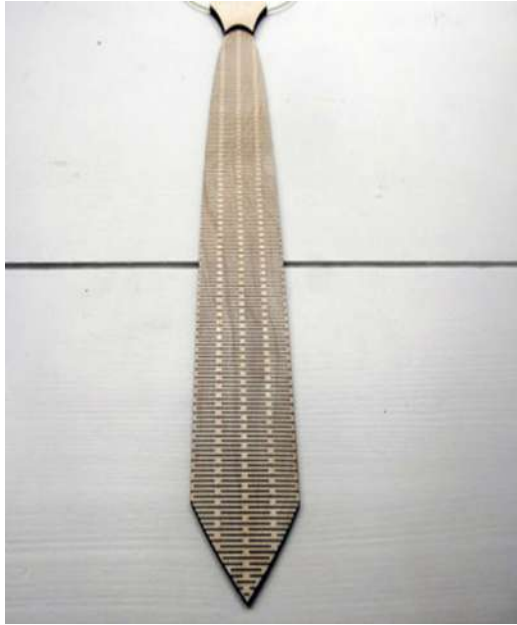
Machine Basics



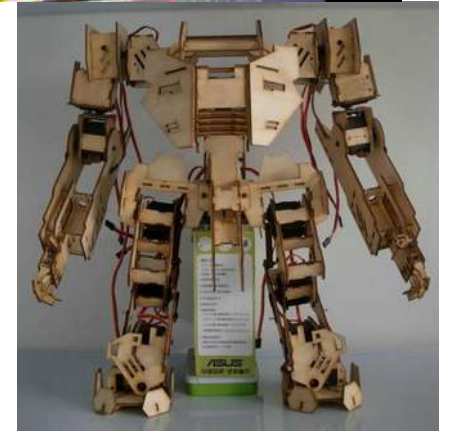
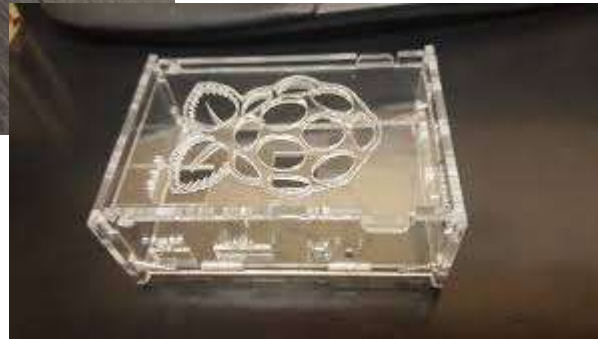
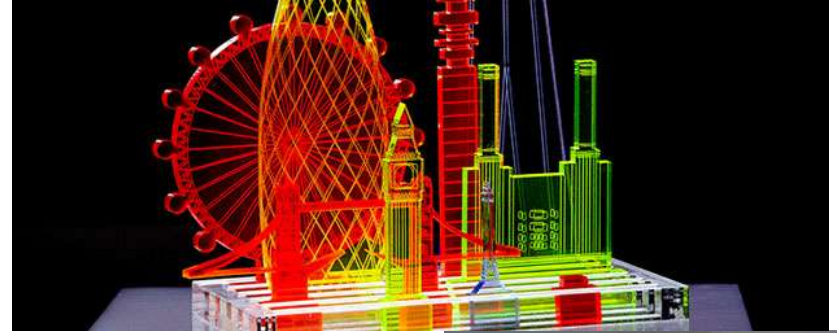
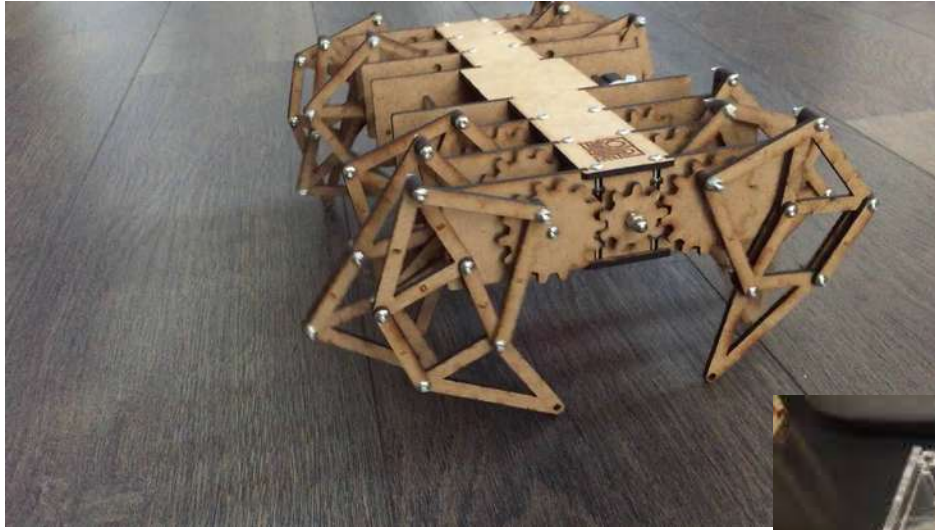
Possible Workflows



Example Projects (Inspiration)



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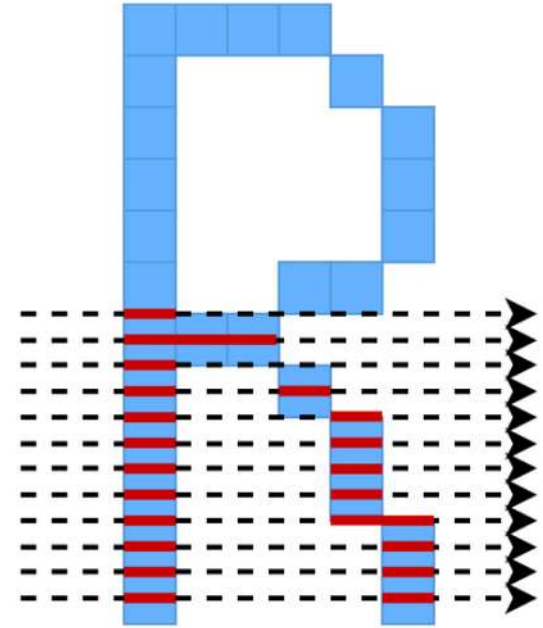
Vector and Raster images

- Vectors are paths that the laser will follow
-
- Raster images are “pixels” or points that will be etched



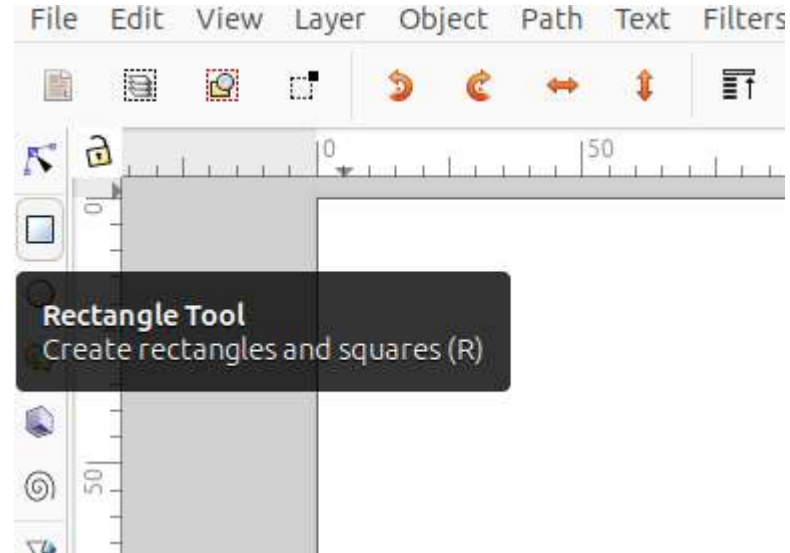
Cutting raster images

- Raster images will be converted to black and white
- Laser head will “Scan” back and forth pulsing the laser for each dot.



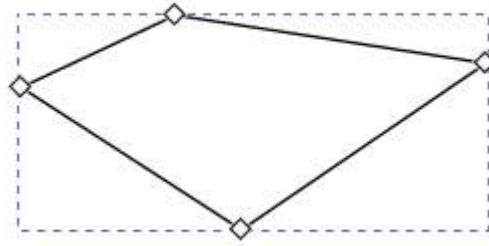
Example design

- Rectangle tool to create rectangles
- Use this to make an outline of your work-piece (cutting board)
- This helps you align your design on the machine relative to what you're engraving.
- You can enter measurements like width and height based on measurements of your work piece.

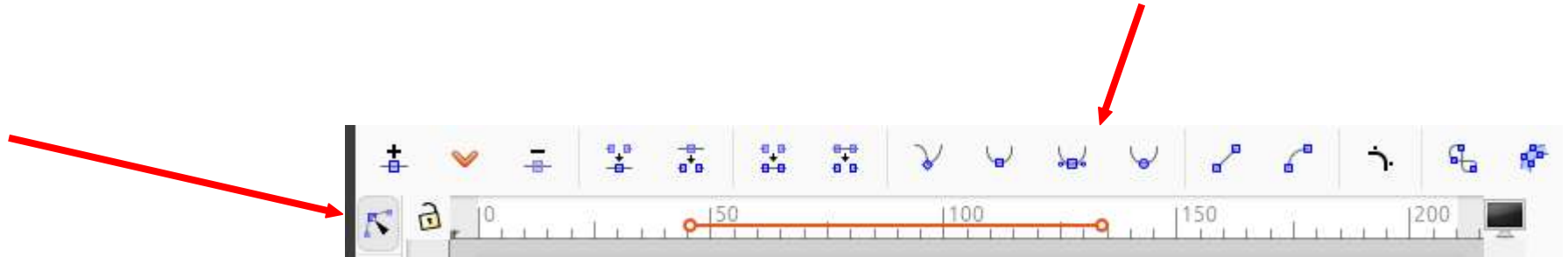


Pencil Tool

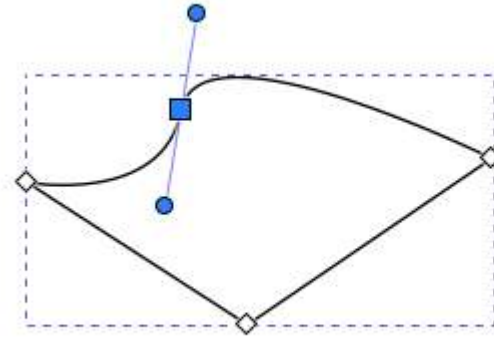
- Pencil tool for straight lines
- Nodes are the endpoints



Nodes

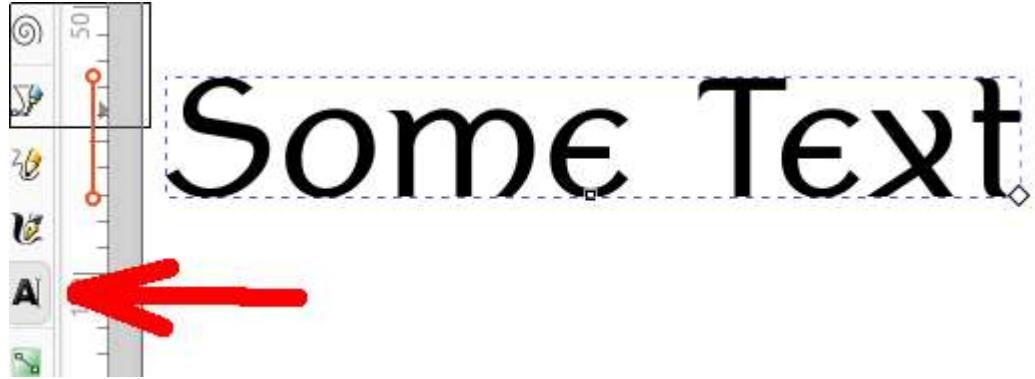


- Click the “Nodes” icon to edit path control points
- Adjust node control points to make curves
- ESCape key to return to object-level editing



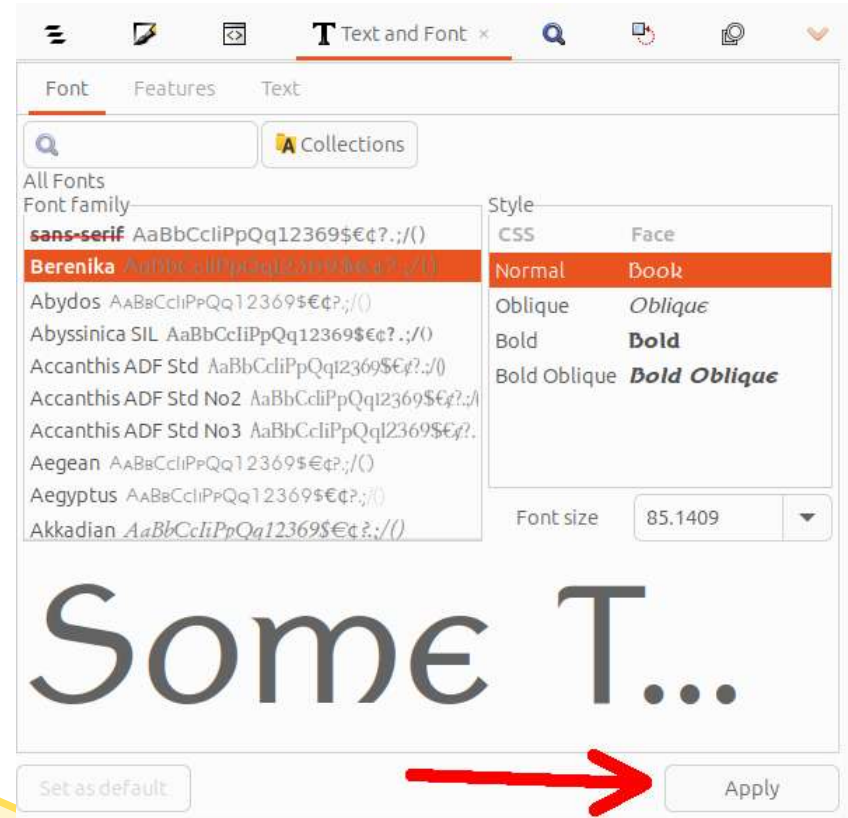
Text Tool

- Incorporate text into your designs
- You can cut it by “outline” or by “filling” to get different effects.



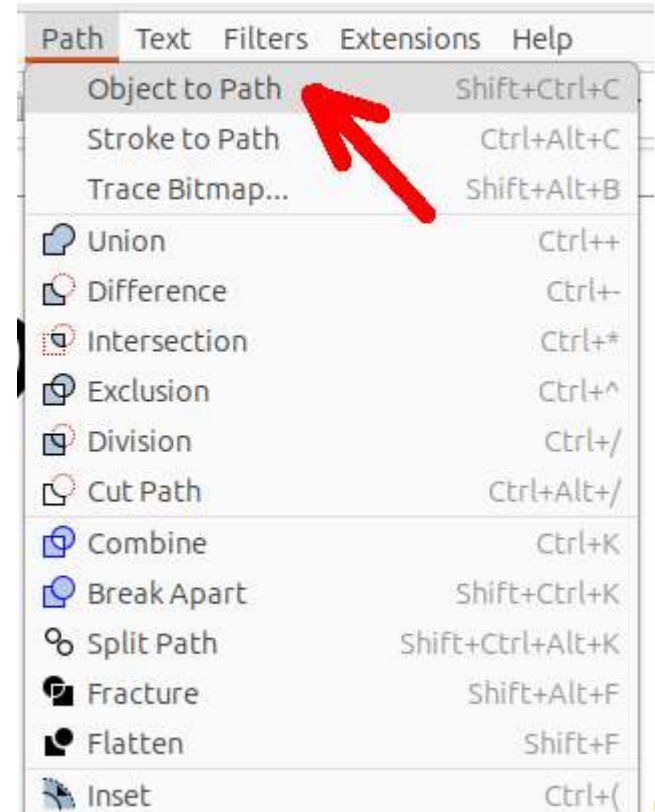
Size and font

- Use the “Text and Font” tab to adjust the look and size of the text.
- Select the text, change it, and hit the “Apply” button.



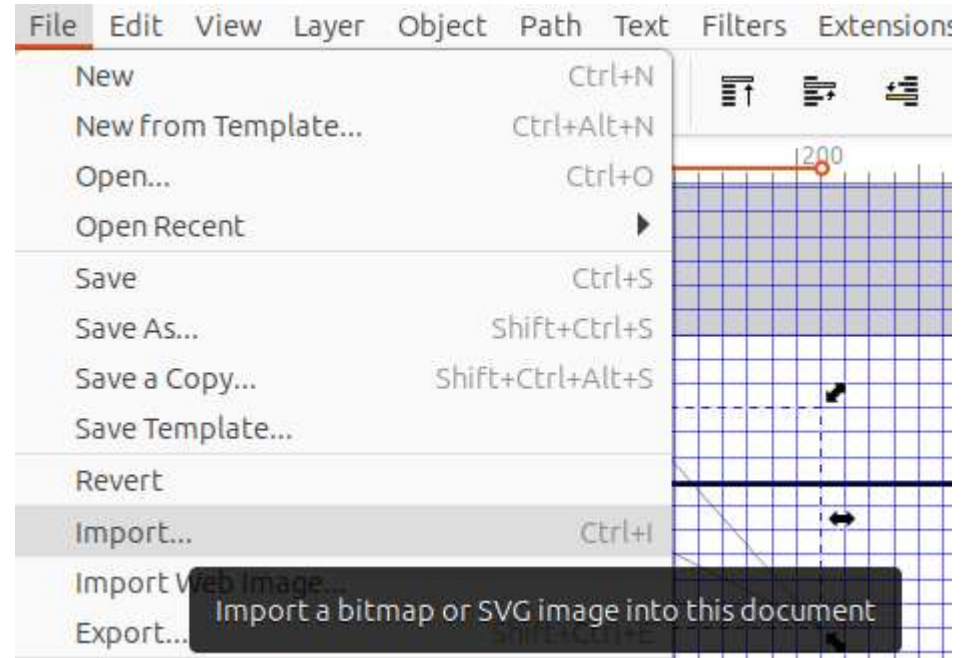
Convert text to paths

- Not all computers have the same fonts.
- To make sure you get the actual font you want, always convert your text to a “path” before you move it over to the machine to cut.
- This takes the text and automatically traces out the path corresponding to it.



Images

- You can also import images into your design
- Or just “Paste” an image directly into your design
- They will only cut “grayscale”, but work pretty well.
- Once imported, you can move and scale it as needed.



Trace bitmap to path

- More advanced technique
- You can “convert” the image into a set of paths for better control
- Somewhat tricky to get the right effect, but can get really good results.
- Once it's a path, you can edit nodes for very fine-grained control
- Works best when there's a lot of contrast in the image



Fill and Stroke

- Set color in fill and stroke section
- Fill = None for just outline
- Fill color set for “engraving” the inside



Cutting Board Dimensions

- Jig locates board at 60mm to the right and 60mm down from the top left corner of the design
- Board measures 256.4mm wide and 383.4mm tall.
- Decide if you want your board to be portrait or landscape
- Make sure your design fits inside the board
- You can use the example design SVG as a starting point



Finished Design

Design starts at 60mm, 60mm

Fits within the cutting board shape

Leaves a little margin at the edges

Now, we'll take your design over to the Lightburn PC to actually cut it.



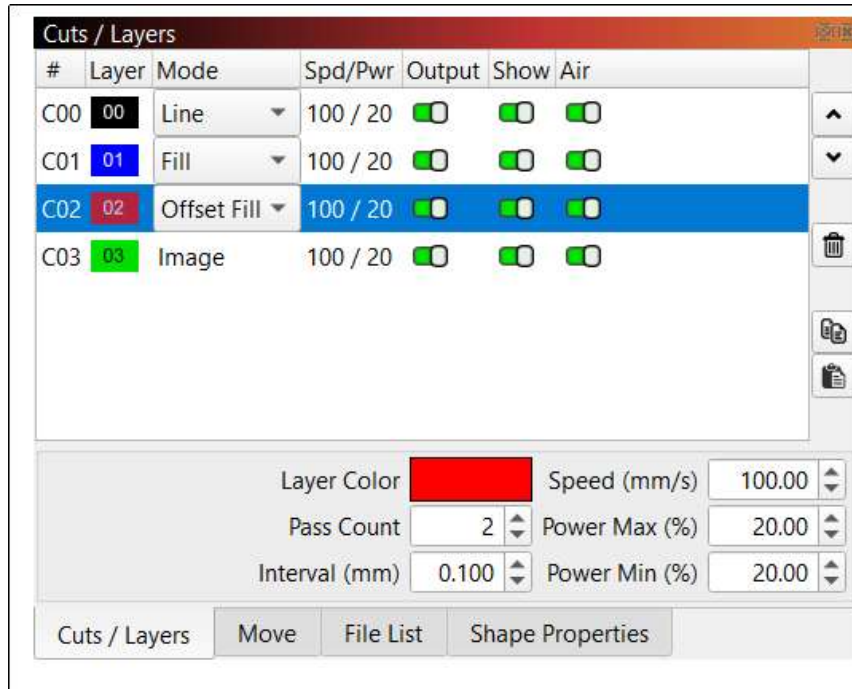
Speed and power

- Each element in your design should be given a “color”. That “color” will later determine how the cuts happen.
- Generally, each color will be assigned a “speed” and a “power” level. This will determine how the material gets cut.
- Generally, higher power = more cutting
- Lower speed = more cutting
- Etch/engrave : High speed, low power
- Cut: Low speed, high power (never above 50%)



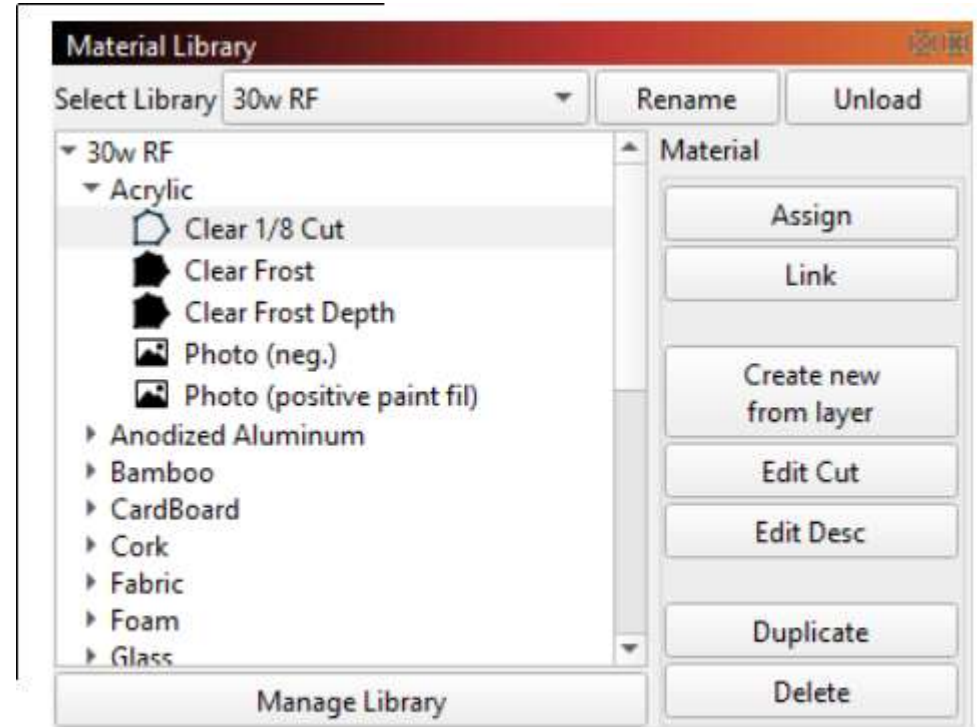
Lightburn

- Import Design
- Set speed and power settings
- Assign each color to a “mode”, and speed/power.
- Never exceed 50% power



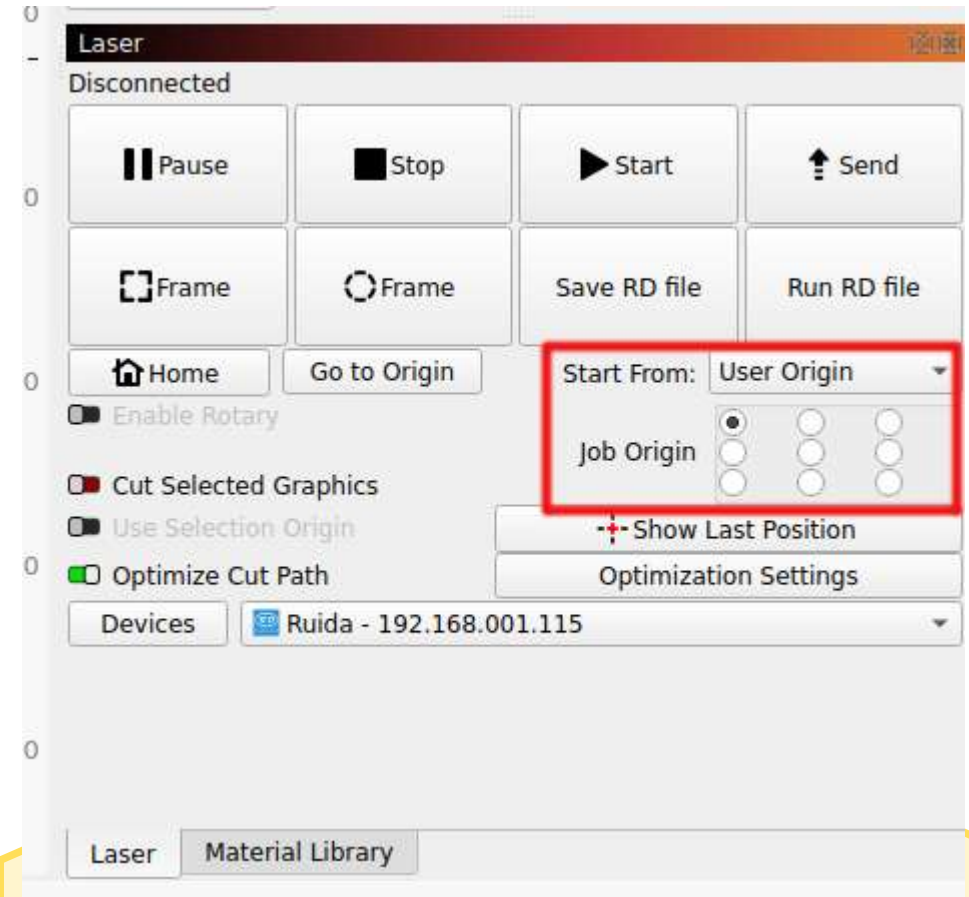
Material Library

- You can set the speed/power manually if you want, or you can “Assign” it from the material library.
- Make sure your “Cut” layer(s) are last, otherwise your work may fall down before it does the engraving layers.



Send to machine for cutting

- Once speed and power are set, we can send our design to the machine to cut it.
- Start button directly starts machine
- Send button sends file to machine. On machine, select file and press start button.
- Since we're using a jig, we're going to use "Absolute" coordinates because we know exactly where the cutting board will be located on the machine.



Safety

- Materials: Approved

- MDF
- Plywood and most woods
- Some plastics
- UNTREATED leather
- Stone, Glass, Ceramic

- Materials: Banned

- Plastics containing chlorine
 - Vinyl, PVC, Nylon
- Resins
- Epoxies
- Metals (Surface engraving is ok, but cutting does not work)



Safety

- Make sure all panels are closed
- Beware of fire and flare-ups on materials. Be prepared to extinguish fire with fire blanket
- Keep lid closed to avoid blindness
- Beware in case lens vaporizes (unlikely but possible), possible Selenium exposure
- Be careful not to touch lenses or mirrors as they can bring the machine out of alignment.
- Persons under 18 must have parent or guardian nearby



Making the cuts

- Put material in machine
- Your supervisor/trainer will turn on the machine, chiller, and air assist as well as opening the air vents.
- Set focal distance between work and laser head.
- Send design to machine using Lightburn
- Close all doors/panels.
- Select File
- Press green “Start/Pause” button.
- Don’t stare at beam
- Don’t leave unattended



Congratulations!

- You have now made your first laser-cut artifact.
- Everything else is creativity and honing your technique

